

“Indian Missile Systems: Origins, Development & Present Status”

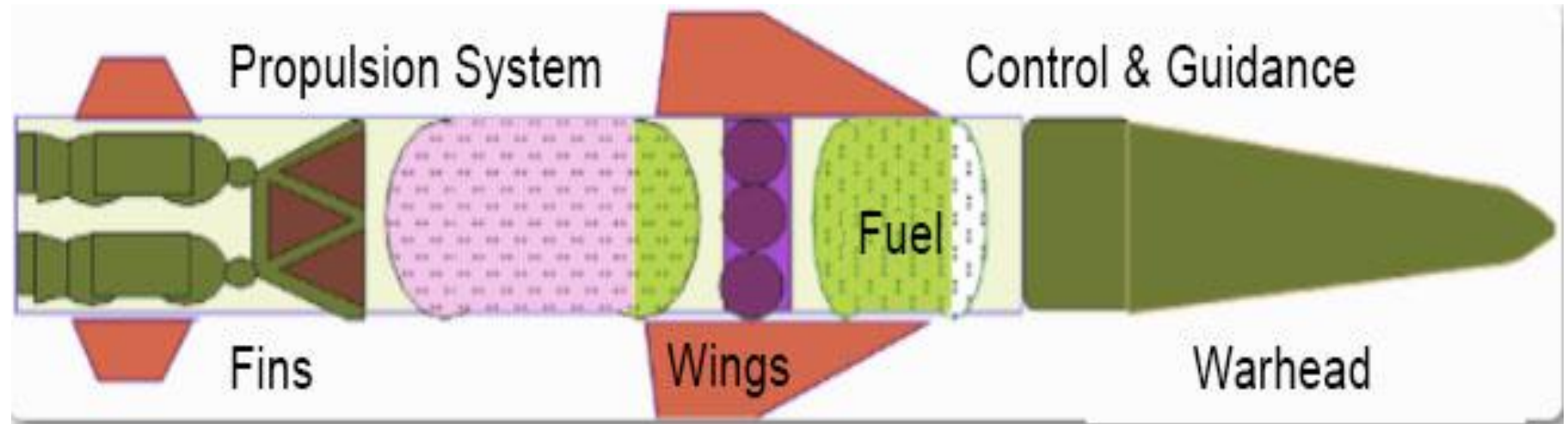


What is a Missile?

A missile is a guided airborne ranged weapon capable of self-propelled flight usually by a jet engine or rocket motor.

Missiles have five system components:

- ❖ Guidance system
- ❖ Targeting system
- ❖ Flight system
- ❖ Engine
- ❖ Warhead



Historical Origins & Early Efforts (1950s–1970s)

- Early attempts (1958 onward) to build missile / rocket technology
- **Projects such as *Project Devil*, *Project Valiant* (1970s) —**
- many got shelved due to technical, organizational constraints
- Learning via collaborations / imported technology vs building domestic capabilities



Launch of the IGMDP (1983)

In 1983, the **Integrated Guided Missile Development Programme (IGMDP)** was launched under DRDO, led by Dr. A. P. J. Abdul Kalam

IGMDP's five missile systems:

- Prithvi (surface-to-surface ballistic)
- Agni (ballistic: medium / long range)
- Akash (surface-to-air)
- Trishul (short-range surface-to-air)
- Nag (anti-tank guided missile)

The program laid foundation for indigenous missile competence

Prithvi (surface-to-surface ballistic)



What are the Prithvi Missiles?

The Prithvi missile system comprises various tactical **Surface-to-Surface Short-Range Ballistic Missiles (SRBM)**.

Its development began in 1983, and it was **India's first indigenous ballistic missile**.

It was first test-fired in 1988, from **Sriharikota**, SHAR Centre.

- It has a range of from 150 to 300 km.

The **naval variant of Prithvi I and Prithvi III class missiles** have the code-name **Dhanush**.

The propulsion technology was likely **based on the Soviet SA-2 surface-to-air missile**.

- **Soviet SA-2 Surface-to-Air Missile:**
 - Developed in the mid-1950s, the Soviet SA-2 surface-to-air missile was the first effective Soviet Union surface-to-air missile.
 - Developed as a missile for the battlefield, it could carry a nuclear warhead in its role as a tactical nuclear weapon.

The Prithvi missile project encompassed developing three variants for use by the Indian Army, Indian Air Force and the Indian Navy. The initial project framework of the Integrated Guided Missile Development Program outlines the variants in the following manner.

Prithvi I (SS-150) –

Army version *(150 km (93 mi) range with a payload of 1,000 kg (2,200 lb))*

Prithvi II (SS-250) –

Air Force version *(350 km (220 mi) range with a payload of 500 kg (1,100 lb))*

Prithvi III (SS-350) –

Naval version *(350 km (220 mi) range with a payload of 1,000 kg (2,200 lb))*

Dhanush (missile)

The Dhanush Missile, also referred to as Prithvi-III, is the naval variant of India's indigenously developed Prithvi-II Missile

Significance

The Dhanush Missile enhances India's maritime strike capabilities in several ways:

Second-Strike Capability:

By being deployable from naval platforms, Dhanush contributes to India's nuclear triad, ensuring a credible second-strike capability.

Versatility:

Its ability to carry various types of warheads makes it adaptable to different mission requirements.

Mobility: Being ship-launched, it provides strategic flexibility and rapid deployment options in maritime scenarios

Dhanush



Agni (ballistic: medium / long range)



Indian ballistic missile series, including Agni-I to Agni-V

Development of Agni Missiles

The Agni missile was developed as part of India's Integrated Guided Missile Development Program (IGMDP) in 1983, which also produced the Prithvi, Nag, Akash, and Trishul missiles.

DRDO originally envisioned a two-stage rocket, the Agni demonstrator for validating re-entry vehicle technologies.

The Agni missile was developed by the DRDO using two existing boosters:

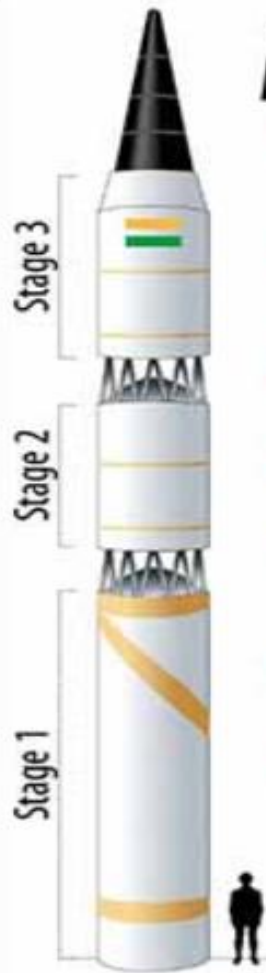
a solid-fueled first stage from India's SLV-3 space launch vehicle and a modified Prithvi-I body for the missile's upper stage.

In 1989, India flight tested the Agni demonstrator for the first time.

The DRDO tested an upgraded Agni demonstrator in 1992, which included a manoeuvring re-entry vehicle and an open interstage, removing the need for ullage motors between the first and second stages. DRDO conducted a third flight of the demonstrator in its upgraded configuration in 1994

AGNI 5 Missile

Nuclear-capable
Inter-continental ballistic
Surface-to-surface



Operational range
5500-5800km

Weight
50000 kg

Length
17.5 m

Diameter
2 m

Speed
24 Mach

Engine
3 stage solid

Warhead
Nuclear
Weight
1500 kg



Successfully
test-fired from
Abdul Kalam
Island off
Odisha coast
on Dec 26,
2016

TESTS
CONDUCTED
RANGE

Agni-1
Apr 19, 2012
700 km

Agni-2
Sep 15, 2013
2000 km

Agni-3
Jan 31, 2015
2500 km to more than 3,500

Agni-1
Nov 9, 2015

Akash (surface-to-air)

THE AKASH SYSTEM	
720 kg Weight	35 cm Diameter
5.78 metre Length	25 km Strike Range
	Mach 2.5 Speed
	18 km Flight ceiling
	60 kg Warhead weight

Akash missile being test-fired from the ITR off the Odisha coast | EXPRESS PHOTO



Akash

It is a Short Range Surface to Air Missile (SAM) system to protect vulnerable areas and vulnerable points from air attacks.

It was indigenously designed and developed by India's DRDO and is produced by Hyderabad-based Bharat Dynamics Ltd. (BDL)

Akash Missile System Features

- ❑ It is a solid-fuel, ramjet propulsion system, capable of reaching speeds of Mach 2.5 to 3.5 (up to 4,200 km/h).
- ❑ It has an operational range of 4.5 to 25 km and can engage targets at altitudes of up to 18 km. The system uses command guidance, a digital autopilot, and includes self-destruct mechanisms to minimize collateral damage in case of target miss.
- ❑ Each missile is capable of carrying a 60 kg warhead, which can be conventional or nuclear. The system boasts a kill probability of 88% with a single missile and up to 99% when fired in a salvo of two.
- ❑ The most important element of the Akash SAM system battery is its high-power, multi-function Rajendra phased array radar

Trishul (short-range surface-to-air)



Nag (anti-tank guided missile)



Types of Missiles Based on Speed

Missiles are classified based on their speed with respect to that of sound, expressed as Mach.

Subsonic missile:

Missiles that travel slower than sound are known as "subsonic."

Examples: U.S. Harpoon anti-ship missile, Indian Prithvi short-range ballistic missile

Supersonic missile:

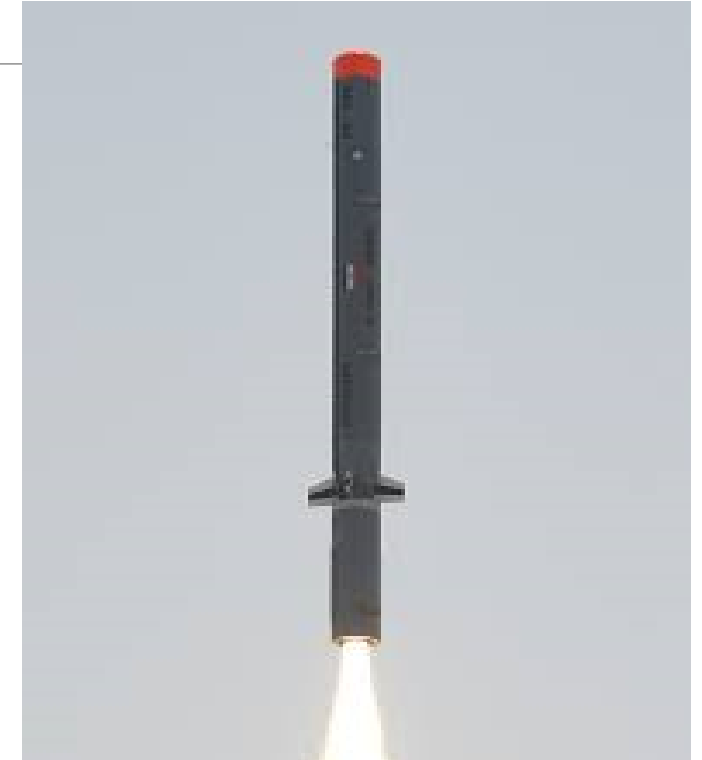
Missiles that travel faster than the speed of sound (Mach 1) but less than Mach 5 are called "supersonic."

Examples: Russian Iskander tactical ballistic missile, Indian BrahMos supersonic cruise missile.

Hypersonic missile:

The term "hypersonic" means that the speed of missiles should be at least five times faster than the speed of sound (over Mach 5). Examples: China's DF-ZF hypersonic glide vehicle, Russia's Avangard, and Shaurya/Sagarika missiles of India (Max speed - 7.5 Mach), etc

Subsonic – Nirbhaya Missiles



Super sonic - Brahmos

Types of Missiles in India Based on Trajectory Ballistic

Missiles:

Ballistic missiles have a ballistic lofted trajectory and work on the principle of unpowered free-fall flight after the boost phase.

Examples are

India's Agni and Prithvi series of ballistic missiles.

Hypersonic Glide Vehicle:

Ballistic missiles attaining hypersonic velocities during the re-entry phase are hypersonic ballistic missiles.

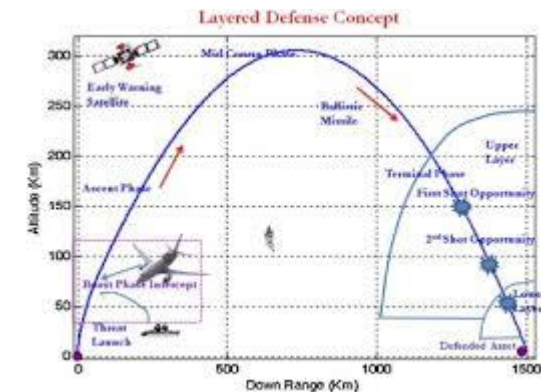
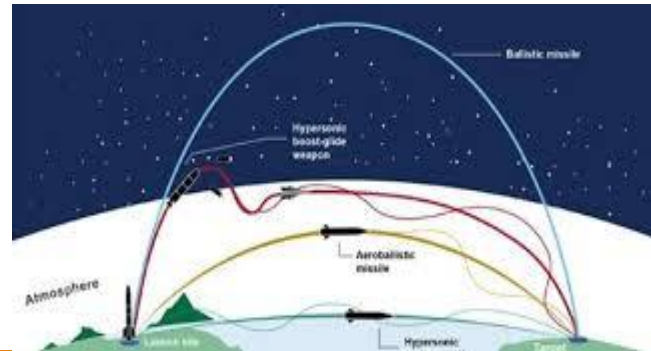
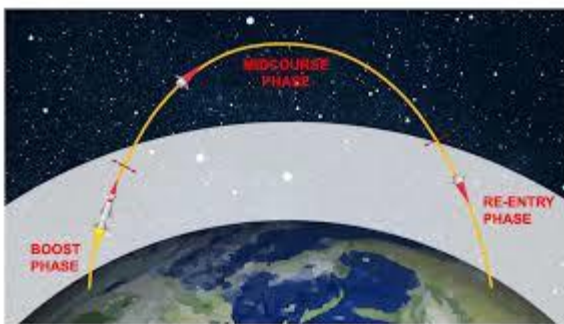
Examples of HGVs: China DF-17 missile mounts hypersonic glide vehicle, Russia Avangard boost-glide system

India is in the “Technology demonstration” stage of the HGVs

A ballistic missile follows a parabolic trajectory by being powered for a short boost phase, then falling unpowered under gravity, reaching high altitudes before descending to the target.

The flight has three phases:

- ❑ a propelled boost phase leaving the atmosphere,
- ❑ a long, unpowered mid-course phase through space or the upper atmosphere, and a final,
- ❑ rapid terminal phase as the warhead re-enters the atmosphere and impacts the target.



DIFFERENCE BETWEEN CRUISE AND BALLISTIC MISSILES



BALLISTIC MISSILE TRAJECTORY

CRUISE MISSILE TRAJECTORY

LAUNCH SITE

TARGET



Based on Strategic Importance

Strategic missiles encompass missiles that give a strategic edge over an opponent or provide credible deterrence to a country.

In India and other powerful countries, these missiles are part of the nuclear triad.

Intercontinental ballistic missile (ICBM):

These missiles are long-range missiles that are capable of delivering nuclear warheads across continents.

It is a part of India's Nuclear Triad.

Agni-V is India's first Intercontinental Ballistic Missile(ICBM), with a range of around 5000-8000 km.

It brings the entire Asia-Pacific region within its reach.

Sub-surface ballistic nuclear (SSBN):

A nuclear-powered submarine carrying and launching ballistic missiles armed with nuclear weapons.

K-15 Sagarika (750 km) and K-4 (3500 km, in development) are the Submarine-launched ballistic Missiles (SLBMs), that complete

India's nuclear triad by enabling second-strike capability from underwater locations.K-15 Sagarika has been deployed in Arihant class SSBN

Based on Tactical Importance

Tactical missiles are employed in the battlefield and frontline combat situations to strike and neutralise enemy assets and capabilities. India boasts an array of tactical missiles.

Prithvi Missiles:

The Prithvi series includes short-range surface-to-surface ballistic missiles. Prithvi-I (150km) and Prithvi-II (350 km) for battlefield roles and hitting strategic targets close to the border. Prithvi-III naval variant (350 km) to boost coastal and seaborne deterrence from warships.

BrahMos Missile:

BrahMos is the fastest supersonic missile of the world.

It is a Mach 3 "fire-and-forget" missile, inducted in all three services of the Indian Armed Forces.

Its variants include land, ship, submarine, and air-launched missiles.

BrahMos II, the hypersonic cruise missile, is currently in the developing phase

Types of Missiles Based on Propulsion

Missiles utilise different types of propulsion and guidance systems based on their range, launch platform and targets. Key missile propulsion systems are:

Solid Propulsion:

It uses solid propellants like Hydroxyl-terminated polybutadiene (HTPB) fuel. It is simple, low-cost, and reliable.

Example: Prithvi, Brahmos

.Liquid Propulsion:

It employs liquid fuels (Hydrazine - N_2H_4 , Liq. hydrogen, etc.) and oxidisers (Nitrogen tetroxide - N_2O_4 , Liq. oxygen, etc.).

It has higher efficiency and throttle ability.

Example: Agni series, Akash.

Recently, India has successfully tested the Agni-Prime missile with solid propellant, making it lighter than other Agni series missiles.

Hybrid Propulsion:

It uses a combination of solid and liquid propellants.

Example:

Used in Brahmos hypersonic cruise missile prototype.

Cryogenic:

It uses liquid oxygen and hydrogen. It has a very high energy density. Example: Employed in long-range Agni-V ICBM

Based on Guidance Systems

Guidance systems enable missiles to accurately reach targets.

Various guidance technologies are:

Command Guidance:

External commands guide missiles through radio/wire links.

Example: Nag anti-tank missile.

Inertial Guidance:

Onboard computers and motion sensors provide autonomous course correction.

Example: Agni ballistic missiles.

Terrain Mapping:

This compares onboard terrain maps to radar altimeter data for accuracy.

Example: Prithvi missiles. Laser

Homing: It guides towards a target illuminated by a laser designator.

Example: Helina anti-tank missile, Anti-Tank Guided Missile for MBT Arjun, etc.

Radar/GPS: This updates position using satellite navigation and matches it to targeting data.

Example: Brahmos cruise missile

Surface-to-Surface	Launched from land-based platforms to strike ground targets	Prithvi, Agni, and BrahMos
Surface-to-Air	Launched from land to intercept aerial threats like aircraft, helicopters, and drones	Akash, MRSAM
Air-to-Surface	Air launched to attack ground-based targets	Helina anti-tank missile fired from Rudra helicopter
Air-to-Air	Launched from aircraft against hostile aerial targets	Astra BVR air-to-air missile fired from Tejas
Ship-to-Ship	Naval anti-ship missiles launched from warships/submarines	BrahMos anti-ship cruise missile fired from ships
Ship-to-Air	Naval surface-to-air missiles for fleet air defence	Barak-8 LR-SAM deployed on ships
Submarine-launched	Fired from submerged submarines	K-15 submarine-launched ballistic missile
Shoulder-fired	Man-portable missiles fired from launchers carried by infantry	FIM-92 Stinger and Igla shoulder-fired SAMs

AIR TO AIR

ASTRA

DISTANCES OF 500 M UP TO 340 KM. ASTRA MK-1 HAS BEEN INTEGRATED WITH INDIAN AIR FORCE'S SUKHOI SU-30MKI.

THE ASTRA MARK 1, HAS A MAXIMUM HEAD-ON LAUNCH RANGE OF 100 KILOMETRES, A SPEED OF 4.5 MACH, AND LAUNCH CLEARANCE UP TO 20 KILOMETRES IN HEIGHT (66,000 FT).

THE ASTRA-1 IS MEANT TO CARRY A 15-KILOGRAM PRE-FRAGMENTED HIGH EXPLOSIVE WARHEAD

[HTTPS://WWW.YOUTUBE.COM/WATCH?V=JPOF2CZFZJO](https://www.youtube.com/watch?v=JPOF2CZFZJO)





ASTRA BVR

IAF-DRDO



Air To Air Missile

NOVATOR KS 172

<https://www.youtube.com/watch?v=LUmlwPtXksQ>







ANTI RADIATION

https://www.youtube.com/watch?v=vJvJ2k_92pl

RUDRAM I

<https://www.youtube.com/watch?v=5FA5oE7AqXQ>





RudraM Series of Missiles

{ RudraM-I, RudraM -II/IIA & RudraM-III }



Pic Credit: Kuntal Biswas



ANTI SATELLITE MISSILES

Prithvi Defence Vehicle Mark II

Tested

operation code named

Mission Shakti

The target of the test was
a satellite present in
a **low Earth orbit**

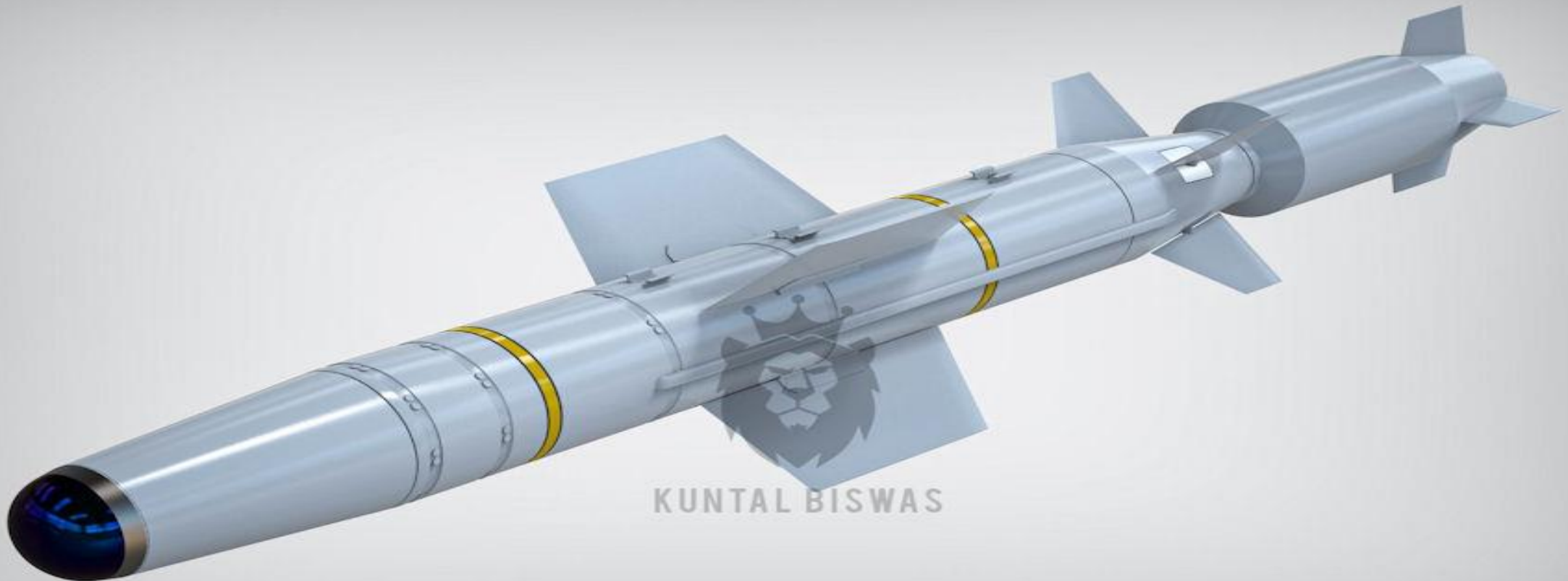




ANTI-SHIP

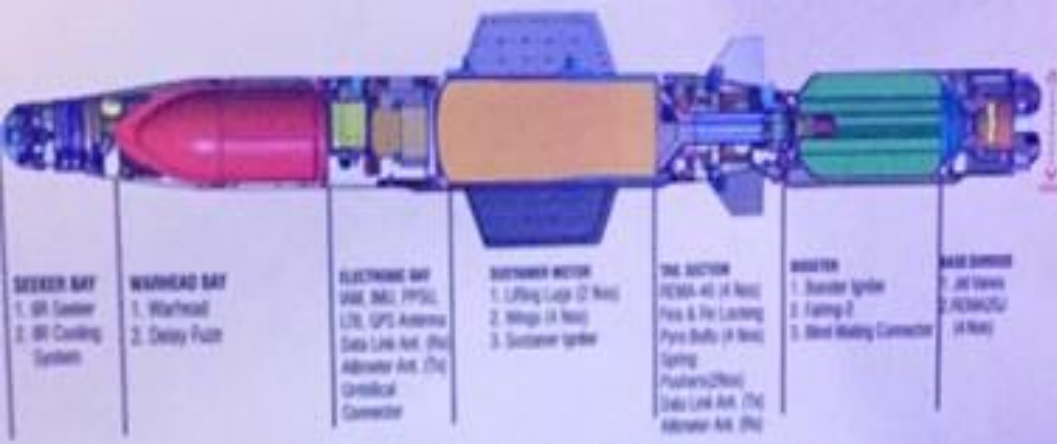
NASM-SR SHORT RANGE ANTI-SHIP MISSILE

<https://www.youtube.com/watch?v=7jQf44H-7CE>



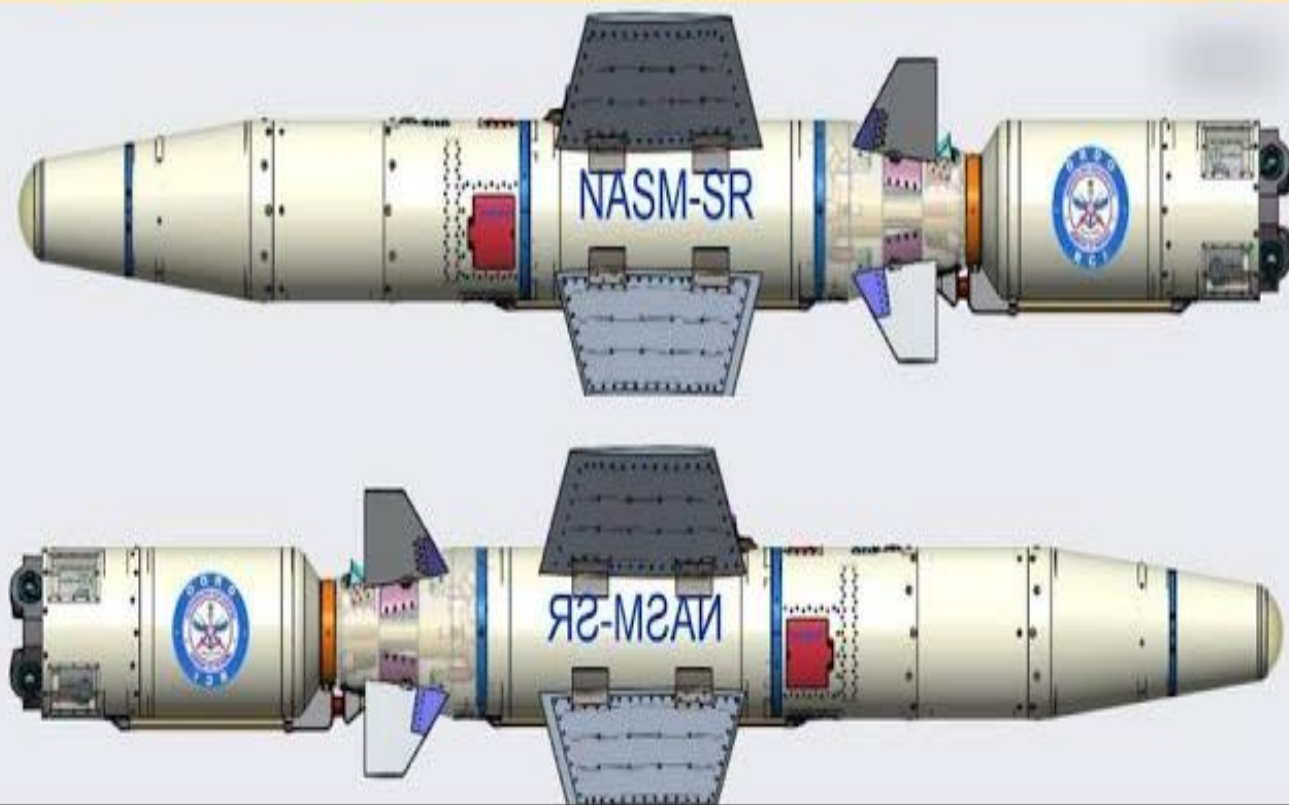
NASM-SR

(NAVAL ANTI-SHIP MISSILE – SHORT RANGE)



DRDO'S NASM-SR

Helicopter Launch Anti-Ship Missile



NAVAL ANTI SHIP MISSILE -MR MEDIUM RANGE ANTI-SHIP MISSILE

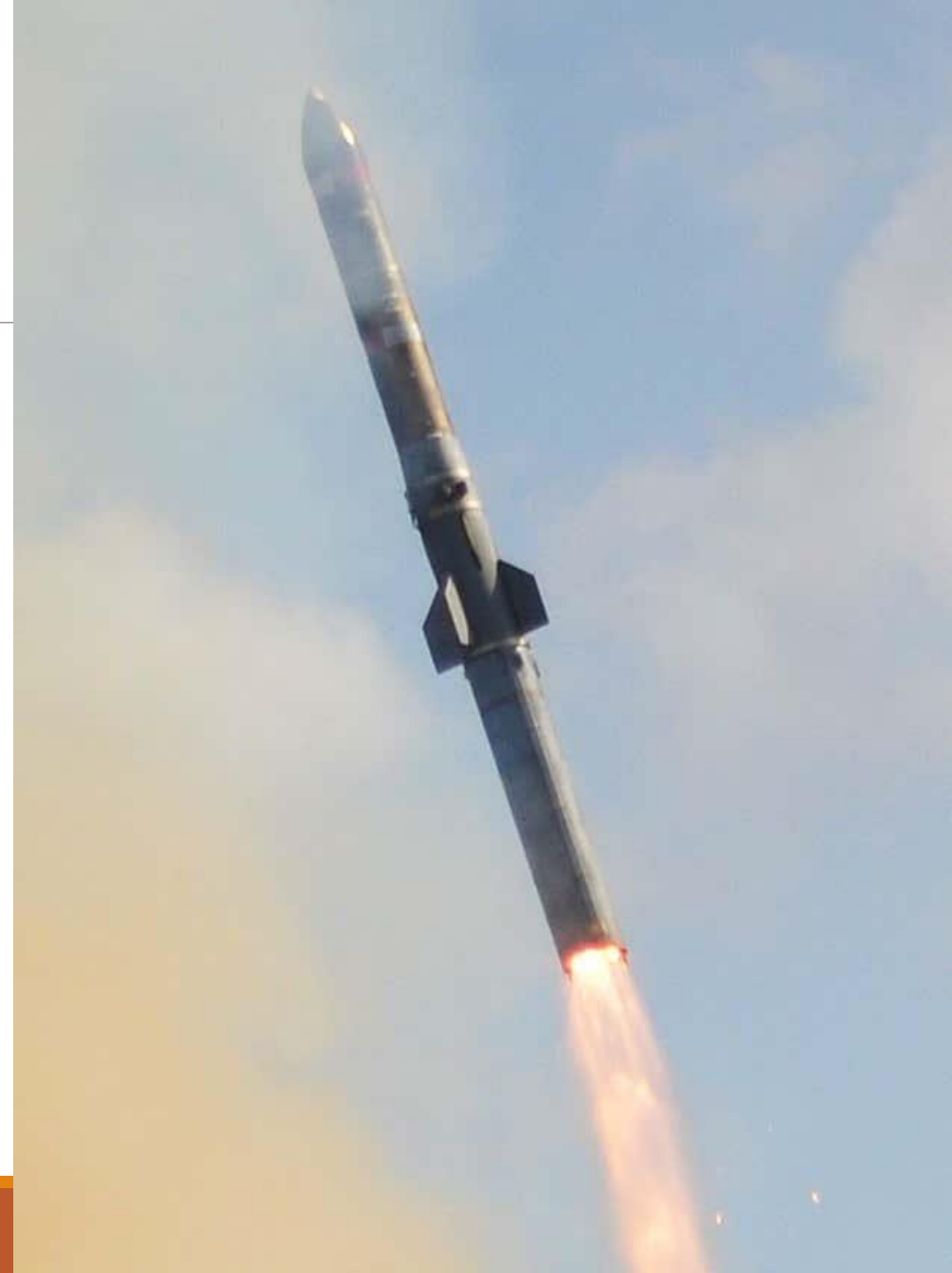








ANTI-SUBMARINE SMART LONG-RANGE ANTI-SUBMARINE MISSILE



Anti-tank

DRDO Anti Tank Missile Wire guided first generation anti-tank missile



ANI



Image tweeted by @ANI

Nag (missile)

Prospina Third generation land-attack ATGM





Nag (missile)
HeliNa/Dhruvastra Air-launched ATGM





Nag (missile)

Stand off Anti-Tank missile Standoff Air-launched ATGM





ANTI TANK

MPATGM







ANTI TANK

Amogha missile

भारत डायनामिक्स लिमिटेड
BHARAT DYNAMICS LIMITED

शांति का आधार अस्त्र-बल
THE FORCE BEHIND PEACE

AMOGHA – III ATGM



APPLICATION:

Amogha-III is a third generation fire-and-forget Anti-Tank Guided Missile.

FEATURES

- Range: 200 – 2500 m.
- Attack mode: Top / direct attack.
- Man-portable.
- Tandem warhead with penetration in excess of 650 mm beyond Explosive Reactive Armor (ERA).
- Dual Mode Imaging Infra-Red (IIR) Seeker.
- Aerodynamic and Thrust Vector Control.
- Smokeless, Signature-free Propulsion System.
- Soft launch.
- Command Launch Unit (CLU) with remote operation capability.



Amogha-1, is a Anti-Tank Guided Missile which has pin point accuracy for a range up to 2.8 km. It is under development by Bharat Dynamics at Hyderabad. It has also become the first missile designed and tested by the Bharat Dynamics Ltd company.

ANTI TANK SAMHO





ANTI TANK

Cannon launched ATGM

blend171@youtube.com

Helicopters



Main Battle Tanks



Armoured Personnel Carriers



Troop Positions

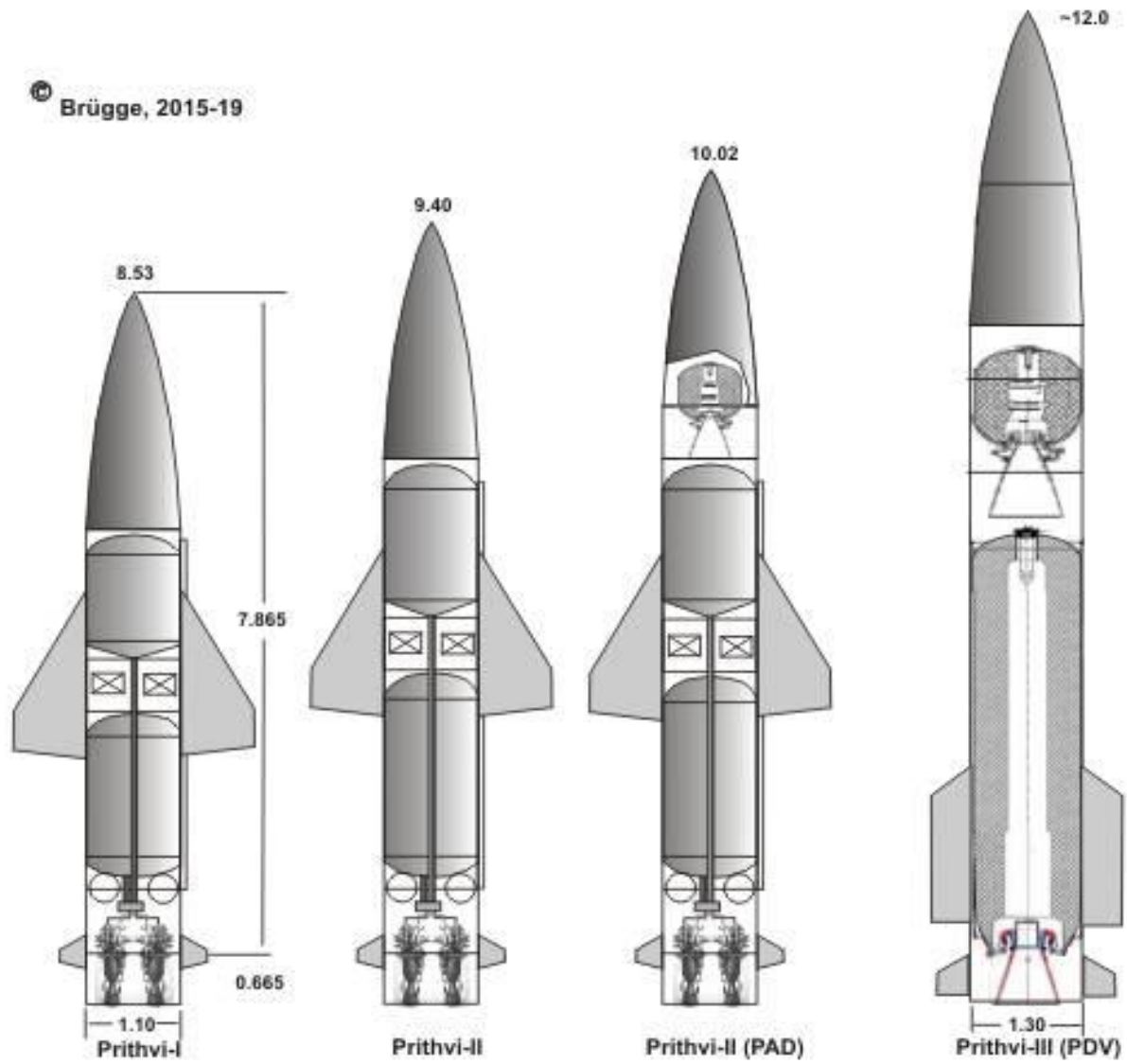


Manportable
AT-5/Milan/FLAME
Launcher

Ballistic



© Brügge, 2015-19





DRUTU 1.2







Dhanush sur



ssile



BALLISTIC MISSILES

Agni (missile)













Classification of Missile Types

By launch mode / domain:

- Surface-to-Surface / Ballistic
- Surface / Air / Sea to Air
- Air-to-Surface
- Submarine-Launched

-
- **By role / target:** ballistic, cruise, anti-tank, anti-radiation, anti-ship, etc.
 - **By speed / trajectory:** ballistic (parabolic), cruise (aerodynamic), hypersonic, etc.

Key Surface-to-Air / Air Defence Missiles

- **Akash system** — medium range SAM, multiple variants
- **Akash-NG (New Generation)**: improved version with dual-pulse solid rocket motor, range ~70–80 km
- **Anant Shastra (QRSAM)** — quick reaction SAM for Army to protect moving columns
- **VSHORAD** (Very Short Range Air Defence)
 - — man-portable / short range, to handle
 - low altitude threats, under development
- **SAMAR** – short-range surface-to-air system
- converting older AAMs like R-73 / R-27 for ground role

Key Cruise, Anti-Radiation & Anti-Tank Systems

- **BrahMos**: supersonic cruise missile (land, sea, air launched)
 - — major joint venture with Russia
- **Nirbhay**: long-range subsonic cruise missile (in development)
- **Rudram**: air-to-surface anti-radiation missile (to target enemy radar)
- **Nag**: anti-tank guided missile, fire-and-forget, many variants
- **Amogha**: ATGM variant
- Anti-ship cruise missiles: e.g., BrahMos (naval version), Klub, etc.
- (from Indian missile template)
